
REQUIREMENTS NOT MET

N/A

PROBLEMS ENCOUNTERED

N/A

HOMWORK EXERCISES

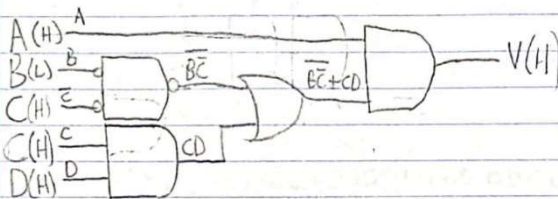
Question 1: Solve the following equations with **any real 1 to 3-input logic gate**, and **any activation level**:

- $V = A * [(B*/C) + C*D]$
- $W = A*B*/C + /(A*B*C)$
- $U = /(A*B + /C)$

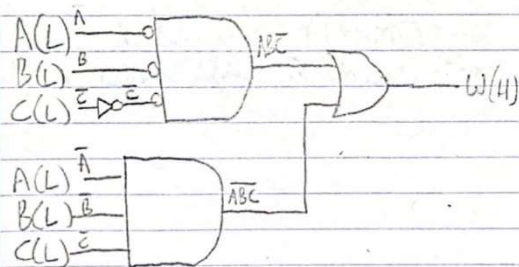
EEL3701 HW 2

1) Any 1 to 3 input gates, minimize gates, any activation

a: $V = A * [\overline{B}C + CD]$



b: $W = ABC + \overline{ABC} = W$



c: $U = \overline{(AB + \overline{C})}$

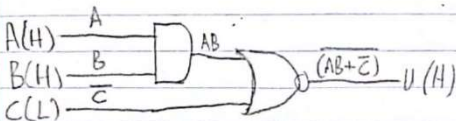


Figure 1: Solutions for 1a through 1c.

Question 2: Solve the following equations with any activation real 1 to 3-input logic gate, **with inputs as active high and outputs as active low**:

- $V = A * [(B*/C) + C*D]$
- $W = A*B*/C + /(A*B*C)$
- $U = /(A*B + /C)$

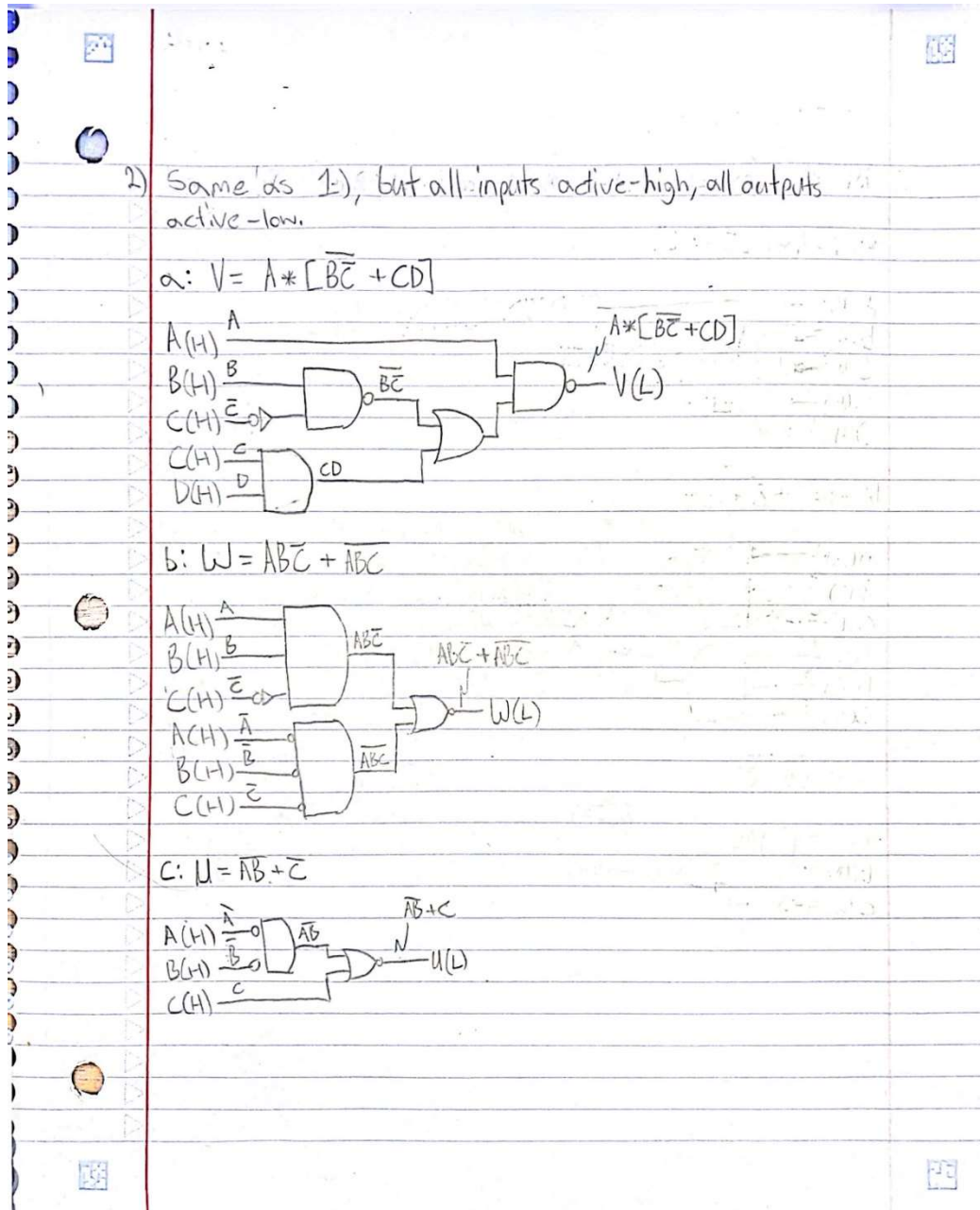


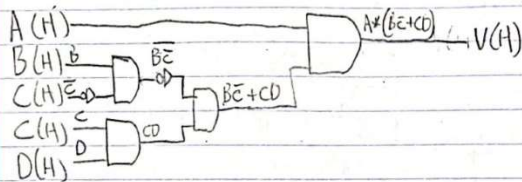
Figure 2: Solutions for 2a through 2c.

Question 3: Solve the following equations with **only AND, OR, Level Shifter** gates or equivalent, assume **all inputs and outputs are active high**:

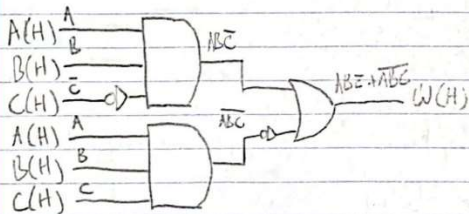
- $V = A * [(B*/C) + C*D]$
- $W = A*B*/C + /(A*B*C)$
- $U = /(A*B + /C)$

3) All AND (BNOR) or OR (BNAND), or Level Shifter, All active-high

a: $V = A * [\bar{B}\bar{C} + CD]$



b: $W = AB\bar{C} + A\bar{B}C$



c: $U = \overline{AB + \bar{C}}$

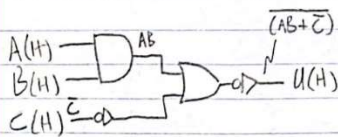


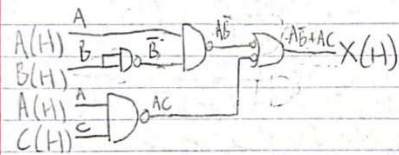
Figure 3: Solutions for 3a through 3c.

Question 4: Design the following logic by direct implementation using only NAND gates or equivalent.

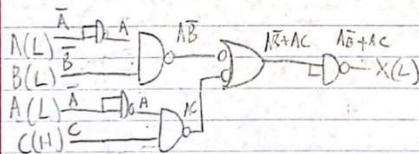
- $X = A \cdot B + A \cdot C$ for $A(H)$, $B(H)$, $C(H)$, $X(H)$
- $X = A \cdot B + A \cdot C$ for $A(L)$, $B(L)$, $C(H)$, $X(L)$
- $X = A \cdot B + A \cdot C$ for $A(L)$, $B(L)$, $C(L)$, $X(L)$

4) Use only NAND or equal

a: $X = A \cdot B + A \cdot C$



b: $X = A \cdot B + A \cdot C$



c: $X = A \cdot B + A \cdot C$

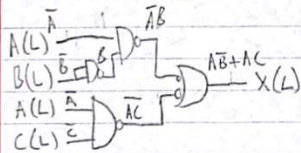


Figure 4: Solutions for 4a through 4c.

Question 5: Create new logic diagrams for problems 1c & 4c, optimizing the number of gates & chips. Label gates and part numbers. Use unused gates on the same chips from one part for the other.

Use activation levels: A(L), B(H), C(H), choose output activation levels.

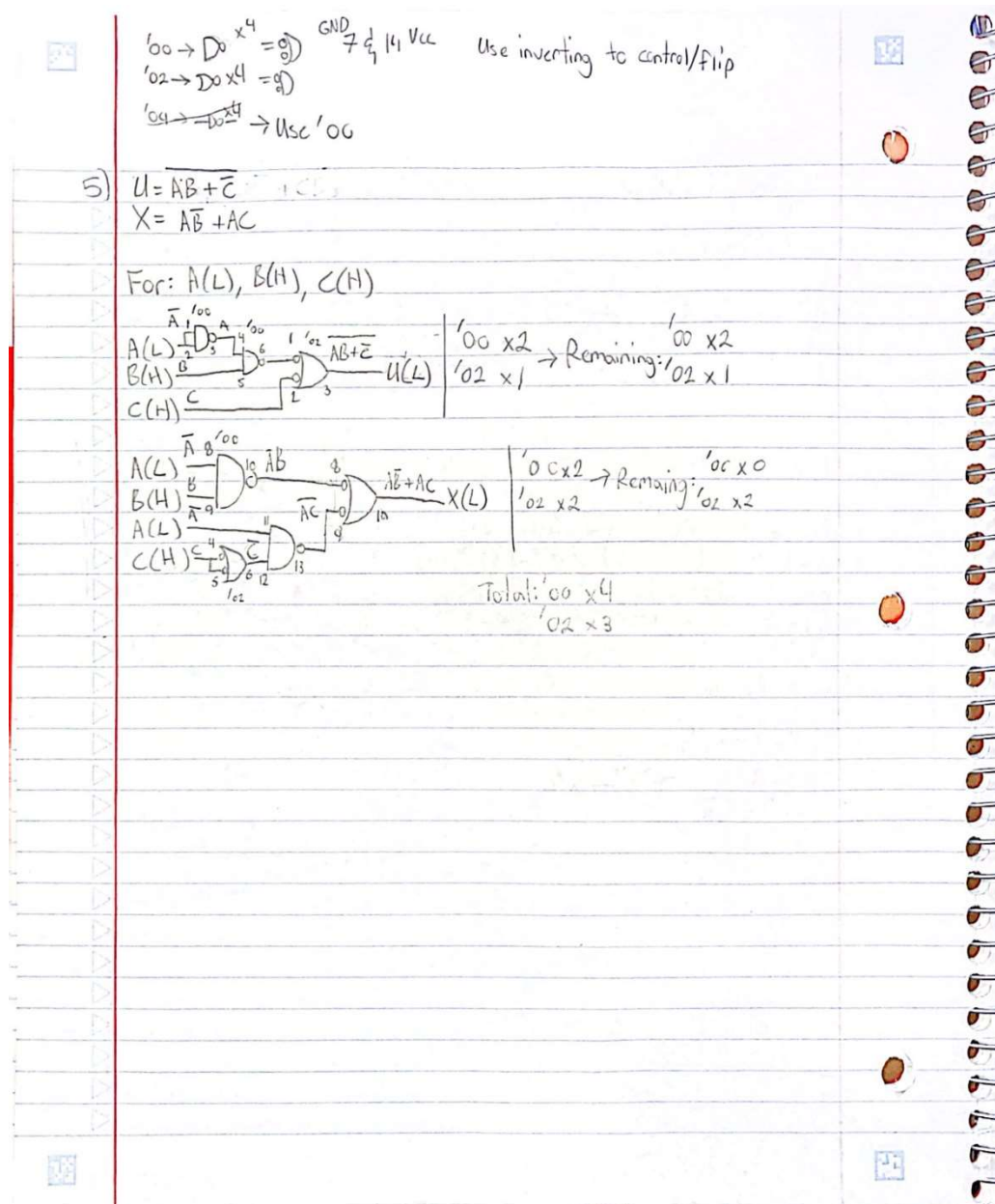


Figure 5: Solution for 5.